DiscoGMPI Mathematical Formulation

DiscoGMPI project

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1 Source-Free Maxwell System

DiscoGMPI advances the source-free three-dimensional Maxwell equations on tetrahedral meshes. On each element K, with scalar positive permittivity ε_K and permeability μ_K, the fields satisfy

$$\varepsilon_K \partial_t \boldsymbol{E} = \nabla \times \boldsymbol{H}, \qquad \mu_K \partial_t \boldsymbol{H} = -\nabla \times \boldsymbol{E}. \tag{1}$$

The electric and magnetic fields are

$$\boldsymbol{E} = (E_x, E_y, E_z), \qquad \boldsymbol{H} = (H_x, H_y, H_z). \tag{2}$$

The material wave speed, impedance, and admittance are

$$c_K = (\varepsilon_K \mu_K)^{-1/2}, \qquad Z_K = \sqrt{\mu_K/\varepsilon_K}, \qquad Y_K = Z_K^{-1}. \tag{3}$$

The divergence constraints are

$$\nabla \cdot (\varepsilon \boldsymbol{E}) = 0, \qquad \nabla \cdot (\mu \boldsymbol{H}) = 0. \tag{4}$$

They are monitored diagnostically through electric and magnetic charge, but they are not advanced as separate equations.

2 Role of the Electric and Magnetic Fields

The Maxwell state is stored as six nodal DG fields. The electric variables determine electric displacement through $\boldsymbol{D} = \varepsilon \boldsymbol{E}$. The magnetic variables determine magnetic flux density through $\boldsymbol{B} = \mu \boldsymbol{H}$. The source-free equations exchange energy between these two partitions through curl coupling.

For the Poisson-bracket path, this partitioning is essential: electric and magnetic unknowns are complementary Hamiltonian variables. The production Poisson-bracket runs use explicit partitioned symplectic Runge–Kutta schemes with the magnetic partition advanced first.

3 Hamiltonian and Poisson Bracket

For homogeneous source-free materials, the continuous electromagnetic Hamiltonian is

$$\mathcal{H}(\boldsymbol{E}, \boldsymbol{H}) = \frac{1}{2} \int_\Omega \left(\varepsilon |\boldsymbol{E}|^2 + \mu |\boldsymbol{H}|^2 \right) \, \mathrm{d}x. \tag{5}$$

Its variational derivatives are

$$\frac{\delta \mathcal{H}}{\delta \boldsymbol{E}} = \varepsilon \boldsymbol{E}, \qquad \frac{\delta \mathcal{H}}{\delta \boldsymbol{H}} = \mu \boldsymbol{H}. \tag{6}$$

The equations can be written as the skew Hamiltonian system

$$\partial_t \begin{bmatrix} \boldsymbol{E} \\ \boldsymbol{H} \end{bmatrix} = \begin{bmatrix} 0 & \varepsilon^{-1}\nabla\times \\ -\mu^{-1}\nabla\times & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{E} \\ \boldsymbol{H} \end{bmatrix}. \tag{7}$$

The corresponding bracket is

$$\{F, G\} = \int_\Omega \frac{\delta F}{\delta \boldsymbol{E}} \cdot \varepsilon^{-1}\nabla \times \frac{\delta G}{\delta \boldsymbol{H}} - \frac{\delta F}{\delta \boldsymbol{H}} \cdot \mu^{-1}\nabla \times \frac{\delta G}{\delta \boldsymbol{E}} \,\mathrm{d}x. \tag{8}$$

The implemented Poisson-bracket DG operator mirrors this skew pairing discretely. Electric volume derivatives use weak derivative matrices, while magnetic volume derivatives use the corresponding transposed operators. Upwind penalties are excluded from the Poisson-bracket formulation because they are dissipative and do not define the implemented partitioned Poisson operator.

4 Weak DG Formulation

Let the domain be partitioned into affine tetrahedra K. On each element, each scalar component is represented in $\mathbb{P}^N(K)$. For vector test functions $\boldsymbol{\Phi}$ and $\boldsymbol{\Psi}$, the elementwise weak equations have the form

$$\int_K \varepsilon \partial_t \boldsymbol{E} \cdot \boldsymbol{\Phi} \,\mathrm{d}x = \int_K (\nabla \times \boldsymbol{H}) \cdot \boldsymbol{\Phi} \,\mathrm{d}x + \int_{\partial K} \mathcal{F}_E \cdot \boldsymbol{\Phi} \,\mathrm{d}s, \tag{9}$$

$$\int_K \mu \partial_t \boldsymbol{H} \cdot \boldsymbol{\Psi} \,\mathrm{d}x = -\int_K (\nabla \times \boldsymbol{E}) \cdot \boldsymbol{\Psi} \,\mathrm{d}x + \int_{\partial K} \mathcal{F}_H \cdot \boldsymbol{\Psi} \,\mathrm{d}s. \tag{10}$$

The minus trace is the trace from the current element, the plus trace is the neighboring or boundary exterior trace, and $\boldsymbol{n}$ is the current element's outward unit normal. The semidiscrete ODE is

$$M_K \dot{U}_K = R_K(U), \tag{11}$$

where M_K is the physical mass matrix and R_K contains volume curl terms and lifted face residuals.

5 Numerical Fluxes

5.1 Centered Flux

For the Hesthaven–Warburton path, define

$$[\![\boldsymbol{E}]\!] = \boldsymbol{E}^+ - \boldsymbol{E}^-, \qquad [\![\boldsymbol{H}]\!] = \boldsymbol{H}^+ - \boldsymbol{H}^-. \tag{12}$$

The centered correction is

$$\mathcal{F}_E^c = \frac{1}{2}\boldsymbol{n} \times [\![\boldsymbol{H}]\!], \qquad \mathcal{F}_H^c = -\frac{1}{2}\boldsymbol{n} \times [\![\boldsymbol{E}]\!]. \tag{13}$$

Centered fluxes are nondissipative in compatible settings and are therefore the natural reference for energy-compatible studies.

5.2 Alternating Flux

An alternating flux selects one trace for one partition and the opposite trace for the complementary partition. Such choices are common in Hamiltonian DG methods because they can preserve a discrete skew pairing when applied consistently. DiscoGMPI currently exposes the Poisson-bracket central/skew operator rather than a separate alternating CLI option. Thus, "alternating" is a DG concept, not an extra production-driver setting.

The implemented Poisson-bracket face corrections are

$$\mathcal{F}_E^{PB} = -\frac{1}{2\varepsilon}\boldsymbol{n} \times (\boldsymbol{H}^- - \boldsymbol{H}^+), \qquad \mathcal{F}_H^{PB} = -\frac{1}{2\mu}\boldsymbol{n} \times (\boldsymbol{E}^- + \boldsymbol{E}^+). \tag{14}$$

5.3 Upwind Flux

The upwind Hesthaven–Warburton flux adds tangential penalties:

$$\mathcal{F}_E^{up} = \mathcal{F}_E^c - \frac{Y}{2}\boldsymbol{n} \times (\boldsymbol{n} \times [\![\boldsymbol{E}]\!])\,, \tag{15}$$

$$\mathcal{F}_H^{up} = \mathcal{F}_H^c - \frac{Z}{2}\boldsymbol{n} \times (\boldsymbol{n} \times [\![\boldsymbol{H}]\!])\,. \tag{16}$$

The penalty terms damp unresolved tangential jumps and introduce numerical dissipation.

6 Boundary Conditions

Boundary conditions are imposed by constructing an exterior trace and then applying the same flux machinery used for interior faces.

6.1 PEC

The perfect electric conductor condition is

$$\boldsymbol{n} \times \boldsymbol{E} = 0. \tag{17}$$

DiscoGMPI uses

$$\boldsymbol{E}^+ = -\boldsymbol{E}^- + 2(\boldsymbol{n} \cdot \boldsymbol{E}^-)\boldsymbol{n}, \qquad \boldsymbol{H}^+ = \boldsymbol{H}^-. \tag{18}$$

6.2 PMC

The perfect magnetic conductor condition is

$$\boldsymbol{n} \times \boldsymbol{H} = 0. \tag{19}$$

The exterior trace is

$$\boldsymbol{E}^+ = \boldsymbol{E}^-, \qquad \boldsymbol{H}^+ = -\boldsymbol{H}^- + 2(\boldsymbol{n} \cdot \boldsymbol{H}^-)\boldsymbol{n}. \tag{20}$$

For cavity validation, the PMC exact mode is obtained by electromagnetic duality:

$$\boldsymbol{E}_{PMC} = Z\boldsymbol{H}_{PEC}, \qquad \boldsymbol{H}_{PMC} = -Z^{-1}\boldsymbol{E}_{PEC}. \tag{21}$$

6.3 Periodic

Periodic boundaries pair two physical boundary faces by translation. The exterior trace on one face is the interior trace from the paired face on the opposite side, with face-node permutation applied so both traces have the same nodal ordering.

The production periodic plane wave is

$$\boldsymbol{E} = \left(0, 0, -\frac{kH_0}{\omega\varepsilon_0}\sin(k(x - x_0) - \omega t)\right), \qquad \boldsymbol{H} = (0, H_0 \sin(k(x - x_0) - \omega t), 0)\,. \tag{22}$$

7 Discrete Diagnostics and Invariants

7.1 Energy

The mass-matrix DG energy is

$$\mathcal{H}_h = \frac{1}{2}\sum_K |J_K| \left[\varepsilon \sum_{a\in\{x,y,z\}} E_{a,K}^T M E_{a,K} + \mu \sum_{a\in\{x,y,z\}} H_{a,K}^T M H_{a,K} \right]. \tag{23}$$

MPI runs include only owned elements locally and use an all-reduce for the global value.

7.2 Charge

The monitored electric and magnetic charges are

$$Q_E = \int_\Omega \nabla\cdot(\varepsilon \boldsymbol{E})\,\mathrm{d}x, \qquad Q_H = \int_\Omega \nabla\cdot(\mu \boldsymbol{H})\,\mathrm{d}x. \tag{24}$$

7.3 Linear Momentum

The monitored momentum density is

$$\boldsymbol{p} = \varepsilon\mu\, \boldsymbol{E}\times\boldsymbol{H}, \tag{25}$$

with total momentum

$$\boldsymbol{P} = \int_\Omega \varepsilon\mu\, \boldsymbol{E}\times\boldsymbol{H}\,\mathrm{d}x. \tag{26}$$

7.4 Angular Momentum

Angular momentum is computed about the coordinate origin:

$$\boldsymbol{L} = \int_\Omega \boldsymbol{x}\times(\varepsilon\mu\, \boldsymbol{E}\times\boldsymbol{H})\,\mathrm{d}x. \tag{27}$$

Changing the coordinate origin changes this diagnostic.

7.5 Optical Chirality

The optical chirality density is

$$\chi = \frac{1}{2}\left(\varepsilon\boldsymbol{E}\cdot(\nabla\times\boldsymbol{E}) + \mu\boldsymbol{H}\cdot(\nabla\times\boldsymbol{H})\right), \tag{28}$$

with global chirality

$$\mathcal{C}_\chi = \int_\Omega \chi\,\mathrm{d}x. \tag{29}$$

8 Error Norms and Rates

Componentwise errors are evaluated by cubature:

$$\|E_a - E_{a,h}\|_{L^2} = \left(\sum_K \int_K (E_a - E_{a,h})^2\,\mathrm{d}x\right)^{1/2}, \tag{30}$$

and similarly for magnetic components. Aggregate errors are Euclidean sums over components. Observed rates use consecutive mesh levels:

$$p = \frac{\log(e_{coarse}/e_{fine})}{\log(h_{coarse}/h_{fine})}. \tag{31}$$

Production convergence checks compare these rates against explicit expected rates instead of checking monotone error decrease.